

Graphing Logarithmic Functions

Algebra 2 / Pre-Calculus Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Identify the domain, range, and intercepts of a logarithmic function
- Understand the relationship between exponential and logarithmic graphs as inverses reflected over $y = x$
- Describe and sketch the behavior of logarithmic functions with varying bases

Problems

1. State the x-intercept of the basic logarithmic function shown below:

$$y = \log_a x$$

2. State the domain and range of the logarithmic function below:

$$y = \log_5 x$$

3. The exponential function below has a y-intercept at (0, 1). What is the x-intercept of its inverse logarithmic function?

$$y = a^x$$

4. Convert the exponential equation below into its equivalent logarithmic form:

$$y = 3^x$$

5. Both functions below are logarithmic. Do they share the same general graph shape and x-intercept? Explain briefly and state the shared x-intercept.

$$f(x) = \log_5 x \quad \text{and} \quad g(x) = \log_{32} x$$



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6. The graph of a logarithmic function is a reflection of its corresponding exponential function over a specific line. Identify that line and write its equation.

7. Evaluate the logarithmic expression below without a calculator:

$$\log_2 8$$

8. Describe how the graph of the function below compares to the graph of $y = \log_7(x)$. Then state the domain and range.

$$y = \log_7(-x)$$

9. Find the inverse of the logarithmic function below, then state the domain and range of the inverse:

$$f(x) = \log_4 x$$

10. Determine the domain, range, x-intercept, and describe the transformation applied to the parent logarithmic function for the function below:

$$y = \log_3(x - 2) + 1$$



Graphing Logarithmic Functions — Answer Key

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Answer Key

1. Answer: (1, 0)

- Set $y = 0$: $0 = \log_a(x)$
- Convert to exponential form: $a^0 = x$, so $x = 1$
- The x-intercept is (1, 0)

2. Answer: Domain: $x > 0$; Range: all real numbers

- The logarithm is only defined for positive values of x
- Domain: $x > 0$ (or $(0, \infty)$)
- The output y can be any real number, so Range: all real numbers (or $(-\infty, \infty)$)

3. Answer: (1, 0)

- The inverse of $y = a^x$ is $y = \log_a(x)$
- Because the inverse swaps x and y coordinates, the y-intercept (0,1) of the exponential becomes the x-intercept (1,0) of the logarithm

4. Answer: $x = \log \text{ base } 3 \text{ of } y$, or $\log_3(y) = x$

- The exponential form $y = 3^x$ means '3 raised to x equals y '
- Converting: $x = \log_3(y)$
- This is the inverse (logarithmic) form of the original equation

5. Answer: Yes, both have the same general shape and share x-intercept (1, 0)

- Both functions are of the form $y = \log_b(x)$ with a positive base
- All such logarithmic functions pass through (1, 0) since $\log_b(1) = 0$ for any valid base b
- The general shape (increasing curve, domain $x > 0$) is the same regardless of the specific base

6. Answer: $y = x$

- Exponential and logarithmic functions are inverses of each other
- Inverse functions are reflections of each other over the line $y = x$
- Therefore the line of reflection is $y = x$

7. Answer: 3

- Ask: 2 raised to what power equals 8?
- $2^3 = 8$
- Therefore $\log_2(8) = 3$

8. Answer: Reflected over the y-axis; Domain: $x < 0$; Range: all real numbers

- Replacing x with $-x$ reflects the graph over the y-axis
- For $\log_7(-x)$ to be defined, we need $-x > 0$, which means $x < 0$
- Domain: $x < 0$; Range: all real numbers (unchanged by reflection over y-axis)



9. Answer: Inverse: $f^{-1}(x) = 4^x$; Domain of inverse: all real numbers; Range of inverse: $y > 0$

- Replace $f(x)$ with y : $y = \log_4(x)$
- Swap x and y : $x = \log_4(y)$
- Convert to exponential form: $y = 4^x$
- The inverse is $f^{-1}(x) = 4^x$, with domain all real numbers and range $y > 0$

10. Answer: Domain: $x > 2$; Range: all real numbers; x-intercept: $(3 - 10^0 \dots$ specifically $x = 2 + 3^{-1} = 2 + 1/3 \dots$ solve: x-intercept at $x = 2 + 1/3 \approx 2.33$); shifted right 2 and up 1

- Parent function is $y = \log_3(x)$, which has domain $x > 0$
- Replacing x with $(x - 2)$ shifts the graph 2 units to the right, so domain becomes $x - 2 > 0$, meaning $x > 2$
- Adding 1 shifts the graph 1 unit upward; Range remains all real numbers
- To find x-intercept, set $y = 0$: $0 = \log_3(x-2) + 1 \rightarrow \log_3(x-2) = -1 \rightarrow x - 2 = 3^{-1} = 1/3 \rightarrow x = 2 + 1/3 = 7/3$
- X-intercept: $(7/3, 0)$; the graph is the parent $\log_3(x)$ shifted right 2 units and up 1 unit

